UNIVERSITY OF EDINBURGH

COLLEGE OF SCIENCE AND ENGINEERING

SCHOOL OF INFORMATICS

INFR11145 TEXT TECHNOLOGIES FOR DATA SCIENCE PG AND INFR11229 UG

Monday 19 th May 2025

13:00 to 15:00

INSTRUCTIONS TO CANDIDATES

1. **Note that ALL QUESTIONS ARE COMPULSORY.**
2. **DIFFERENT QUESTIONS MAY HAVE DIFFERENT NUMBERS OF TOTAL MARKS. Take note of this in allocating time to questions.**
3. **This is a NOTES NOT PERMITTED, CALCULATORS PERMITTED examination.**

   Notes and other written or printed material MAY NOT BE CONSULTED during the examination.

   CALCULATORS MAY BE USED IN THIS EXAMINATION.

Year 4 Courses

Convener: M.Cryan
External Examiners: D.Parker, A.Pocklington, R.Mullins

THIS EXAMINATION WILL BE MARKED ANONYMOUSLY

Please provide answers to **ALL** of the following questions. It is more important to have a clear answer than a longer answer. When calculations are needed, please **show steps** and give answer to **three** decimal digits.

1. (a) For each of the following preprocessing steps, explain the step, and list an advantage and a drawback with an example of each situation: *[3 marks]*

   i. Stopping

   ii. Removing URLs (e.g. URL in a tweet text)

   (b) "With the advent of LLMs, web search engines are no longer needed". Respond to the previous statement by agreeing or disagreeing while discussing your justifications *[2 marks]*

   (c) In the TFIDF formula, log_{10} is usually used in calculating the tf and df components. If someone decided to replace it with log_2:

   i. How should this impact the values of TF and IDF? *[2 marks]*

   ii. How do you think this can affect the ranking of documents? *[2 marks]*

   (d) Regarding Heap's law, which is expected to have smaller b value, a closed-domain collection or a general-domain collection? Why? *[2 marks]*

   (e) System X retrieved a set of documents to a given query, where the relevance of top 5 ranked documents are: $\{2, 1, 3, 0, 3\}$ in order.

   i. Assume the collection have many highly relevant documents with relevance 3, calculate nDCG@5. *[2 marks]*

   ii. Assuming only documents with relevance 3 to be considered relevant, calculate MRR. *[1 mark]*

   (f) Spiders crawl webpages to build an index for search engines. However, some websites get hacked, replacing their original content with irrelevant content.

   i. Why this is a problem? *[1 mark]*

   ii. What key mechanism allows spiders to detect and mitigate this issue? *[2 marks]*

   (g) You have a large-scale Retrieval-Augmented Generation (RAG) system with query and document encoders that generate embeddings for documents. You are tasked with improving the system by updating the encoders. Why it is NOT practical to update the document encoder? *[3 marks]*

2. The Parliament in a given country has decided to record all their sessions and make them searchable. They have hired you as an IR expert to build the engine for them. You have noticed that the automatic speech recognition (ASR) systems available in the market for their language have poor performance (50% word error rate). However, you have also noticed that if you make the ASR system produce the most likely words for each audio segment, there is a high chance (90%) of finding the correct word among them. Thus, you have decided to design a new *non-deterministic* inverted index that enables indexing the ***three*** most likely words for each audio segment along with their probabilities. Below is an example of the ASR output that corresponds to the sentence "`creating jobs is our top priority`", where each term in the document is represented as a vector of the most likely words along with their probabilities.

w_1		w_2		w_3		w_4		w_5		w_6	
treating	0.4	**jobs**	0.35	**is**	0.7	hour	0.55	**top**	0.45	security	0.3
mitigating	0.3	hobs	0.3	oz	0.2	**our**	0.3	good	0.4	minority	0.3
deleting	0.15	dobs	0.25	ezzy	0.05	awa	0.1	job	0.1	**priority**	0.25
creating	0.15	ops	0.1	it	0.05	out	0.05	drop	0.05	seniority	0.05

(a) Please propose a design of a positional inverted index that can index this document representation. Illustrate your answer by giving examples of what the posting list would look like. [*4 marks*]

(b) Would proximity search using your proposed approach work the same was as it does with a standard index (with deterministic terms)? [*1 mark*]

(c) Explain how tf, df, and cf can be calculated in this situation, and how each is different from the normal situation (when terms are deterministic). [*2 marks*]

(d) Suppose the above phrase (w1 to w6) is part of a document with no other occurrences of these terms in the document. What is the tf score of the query "Job creation" after applying standard preprocessing and ignoring IDF? [*1 mark*]

(e) How do you think the performance of this system compares to a standard system –where only most probable recognised term is indexed– in terms of precision and recall? [*2 marks*]

3. You have listened to music. A lot of music. You would like to know how often certain words appear in certain types of songs. For simplicity, you have assigned each song to exactly one genre, such as *pop* or *rock*. You have identified a number of words that are of interest to you, such as *moon* and *tears*, and counted how many songs from each genre they appear in:

	Pop	Rock	Hip-hop	Country	Jazz	Electronic
Total no. of songs	403	1243	756	1202	341	1272
love	85	602	43	769	55	132
night	92	48	337	25	71	115
dance	123	52	24	176	68	422
heart	78	113	44	84	153	202
dream	182	23	15	67	95	182
moon	98	93	12	15	102	152
tears	57	83	105	5	79	38

Calculate the expected mutual information $I(U;C)$ for the word *heart* and the genre *hip hop* (show steps).

$$I(U;C) = \sum_{e_t \in \{1,0\}} \sum_{e_c \in \{1,0\}} P(U = e_t, C = e_c) \log_2 \frac{P(U = e_t, C = e_c)}{P(U = e_t)P(C = e_c)}$$

$$= \frac{N_{11}}{N} \log_2 \frac{NN_{11}}{N_{1.}N_{.1}} + \frac{N_{01}}{N} \log_2 \frac{NN_{01}}{N_{0.}N_{.1}} + \frac{N_{10}}{N} \log_2 \frac{NN_{10}}{N_{1.}N_{.0}} + \frac{N_{00}}{N} \log_2 \frac{NN_{00}}{N_{0.}N_{.0}}$$

[*5 marks*]

4. Provide a short answer to each of the following questions:

   (a) The word “latent” means “hidden” or, in the context of Latent Dirichlet Allocation “not (directly) observable”. What does it refer to? Describe briefly what is not observable (latent), and what is observable in LDA.

   (b) Describe briefly a situation where it makes sense to use a sentiment dictionary instead of a machine learning model.

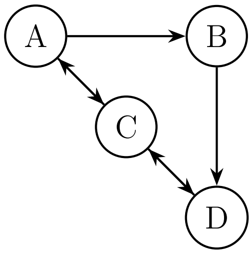
   (c) Where do the topic labels in topic modelling (using LDA) come from?

   (d) In classification, what does “multi-class” mean, and what does “multi-label” mean?

   (e) Describe the difference between a test set and a validation test. When do we use them and why do we need both?

[*5 marks*]

5. Consider the directed graph below. Apply the PageRank algorithm without teleportation. That is, the random surfer never jumps to a random node in an iteration. Answer the questions below. Show your work.



   (a) What are the rankings of nodes (descending) after the 1st iteration? *[2 marks]*

   (b) What are the rankings of nodes (descending) after the 2nd iteration? *[2 marks]*

   (c) Explain why we should or do not need to employ teleportation? *[1 mark]*

6. You are designing a coding assistant named ElmasAI, which is a Retrieval-Augmented Generation (RAG) system that integrates with a user's development environment (e.g., Visual Studio). It has access to local files on your computer. It employs OpenAI's GPT model through API calls.

   ElmasAI provides real-time feedback on code mistakes and can generate code upon request. It should perform well in projects with multiple code files deployed locally.

   Answer the following questions:

   (a) Describe the potential data and data sources to train and fine-tune ElmasAI *[1 mark]*

   (b) Apart from the data that is used to train and fine-tune ElmasAI, what are the documents ElmasAI should use to create the vector database upon integration? *[2 marks]*

   (c) Your users have security and privacy concerns because ElmasAI requires access to the internet to work! How could you modify ElmasAI to address this concern for good? What is your proposed solution's drawback? *[2 marks]*